

Q5. binary search**CODE**

```
def rotated_binary_search(arr, target):
    low = 0
    high = len(arr) - 1

    while low ≤ high:
        mid = (low + high) // 2

        if arr[mid] == target:
            return mid

        # Left half is sorted
        if arr[low] ≤ arr[mid]:
            if arr[low] ≤ target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1

        # Right half is sorted
        else:
            if arr[mid] < target ≤ arr[high]:
                low = mid + 1
            else:
                high = mid - 1

    return -1

data = [4, 5, 6, 7, 0, 1, 2]
t = int(input("Target: "))
result = rotated_binary_search(data, t)
print("Found at:", result)
```

OUTPUT

Target: Found at: 4